

Oral Diagnosis and Management of Patients with Hematologic Cancers

Integrated Biomedical and Clinical Sciences: OMFS 513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Outcomes & Competency Statements

Learning Outcomes

1. Review the physiology, pathology, clinical and oral manifestations of hematologic cancers.
2. Discuss the modalities used in the diagnosis and treat hematologic cancers.
3. Understanding how systemic disease can affect dental treatment.
4. Working the medical team to ensure medically compromised patients receive the safest treatment.

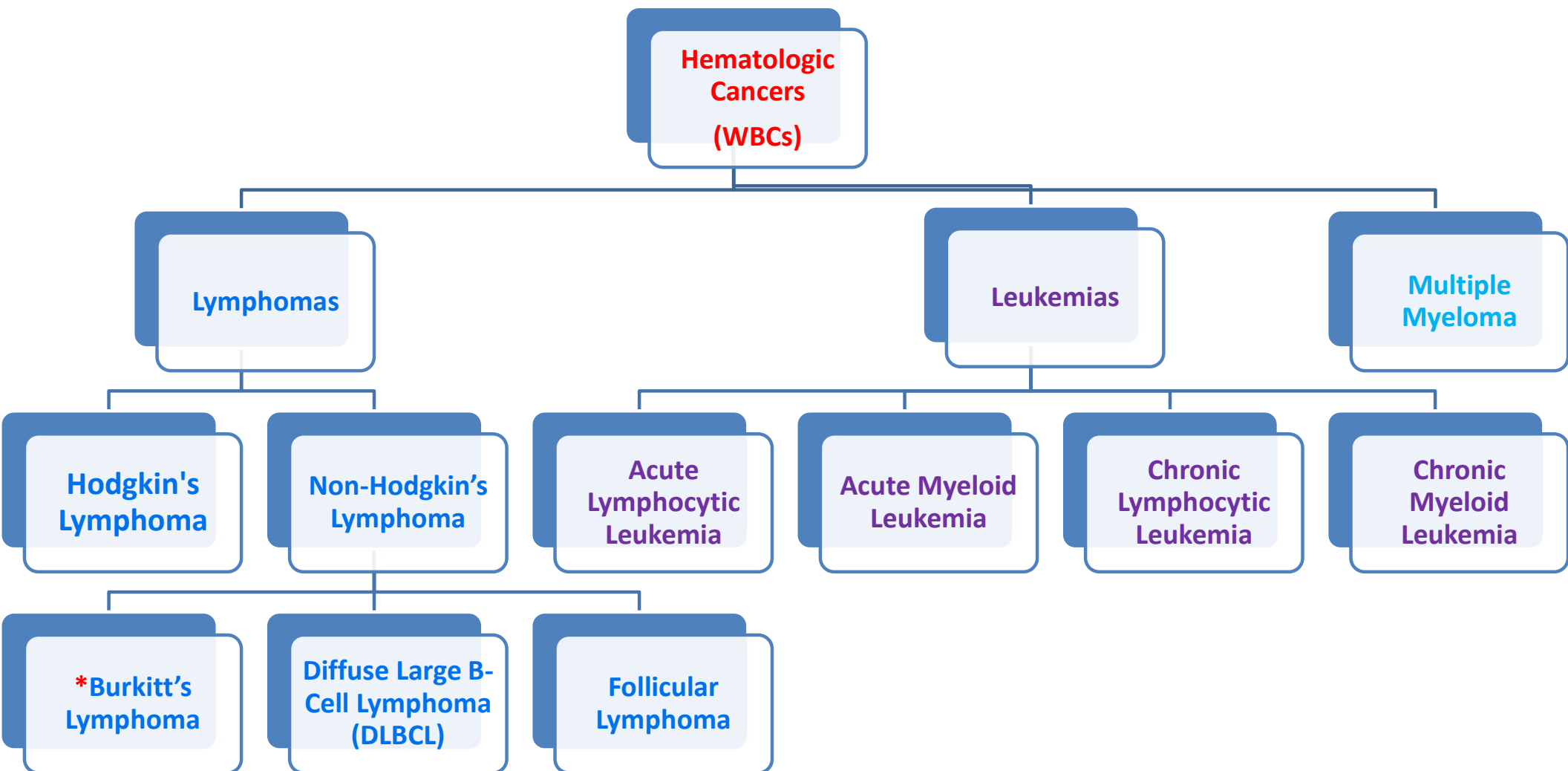
Competency statements met

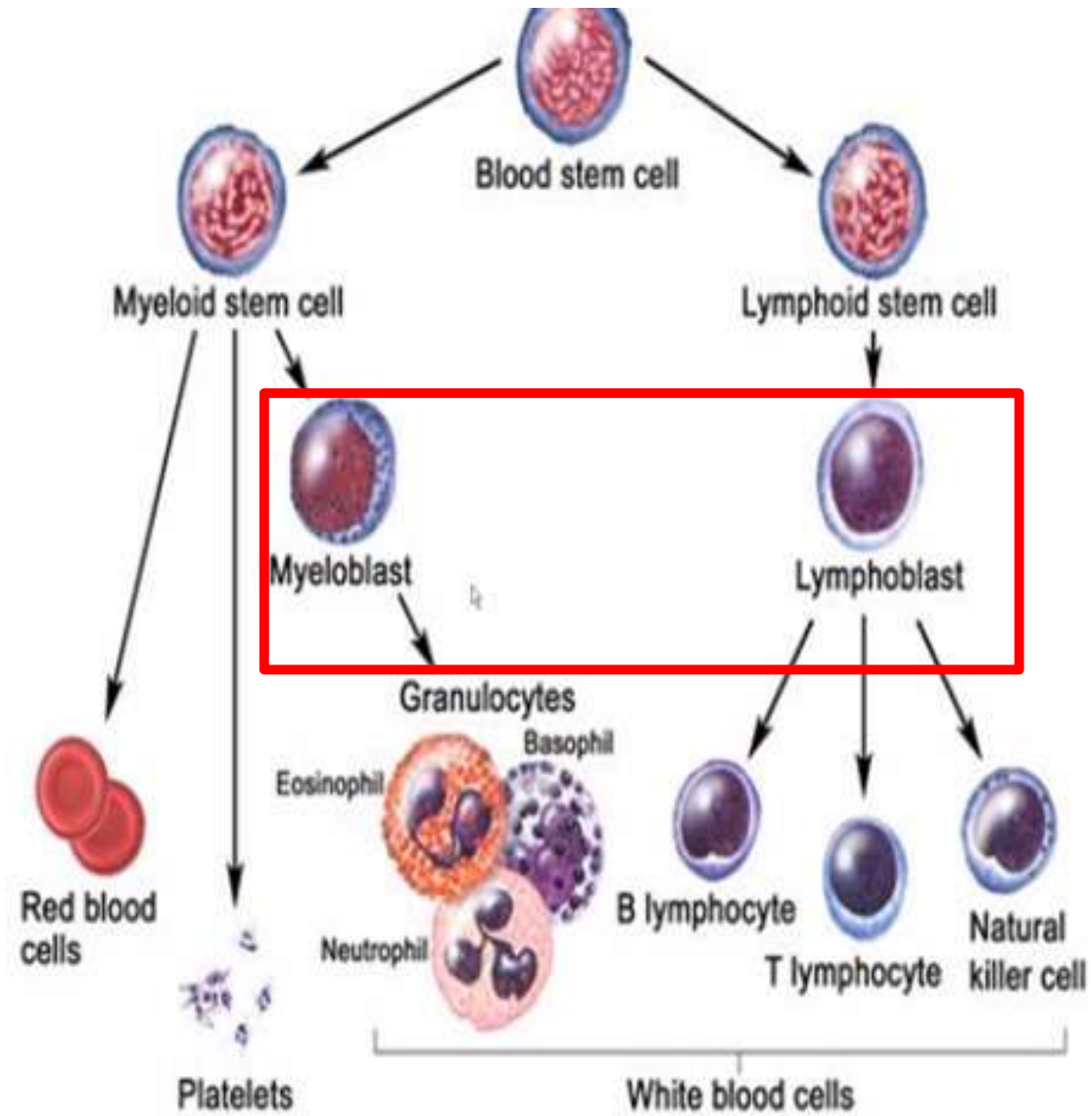
1.3, 2.1, 3.1, 4.4, and 5.1

Hematologic Cancers

- A heterogenous groups of diseases affected millions of people globally each year
- 7% of newly diagnosed cancers
- Leukemias accounts for 2.5% of all diagnosed cancers
- 80,000 Americans are diagnosed with lymphoma each year
- Multiple myeloma is rare hematologic cancer

[Auberger, Patrick; Tamburini-Bonnefoy, Jerome; Puissant, Alexandre.](#) [International Journal of Molecular Sciences; Basel](#) [Vol. 21, Iss. 17,](#) (2020): 6091.





Hodgkin's vs Non-Hodgkin's Lymphoma

Table 23.6

Comparison of Hodgkin and Non-Hodgkin Lymphomas

Parameter	Hodgkin Lymphoma [HL]	Non-Hodgkin Lymphoma [NHL]
Cellular derivation site	B cell	B cell>T cell or NK cell
Localized	Common	Uncommon
Waldeyer ring	Rarely involved	Commonly involved
Extranodal	Uncommon	Common
Abdominal (mesenteric nodes)	Uncommon	Common
Mediastinal	Common	Uncommon
Bone marrow	Uncommon	Common
"B" symptoms (fever, night sweats, weight loss)	Common	Uncommon
Curability	More favorable than NHL	Less favorable than HL

NK, natural killer.

Hodgkin's Lymphoma (HL)

- Bimodal age distribution
 - Most common cause of lymphoma in young adults
 - Higher incidence in males
 - Cause can be associated with viral infections (Epstein-Barr Virus)
 - 3 types based on pathologic features:
 - Classic- 4 subtypes
 - Nodular lymphocyte predominant
 - HL, not otherwise classifiable
 - Key histologic feature : Reed-Steinberg Cell “owl-eyed”



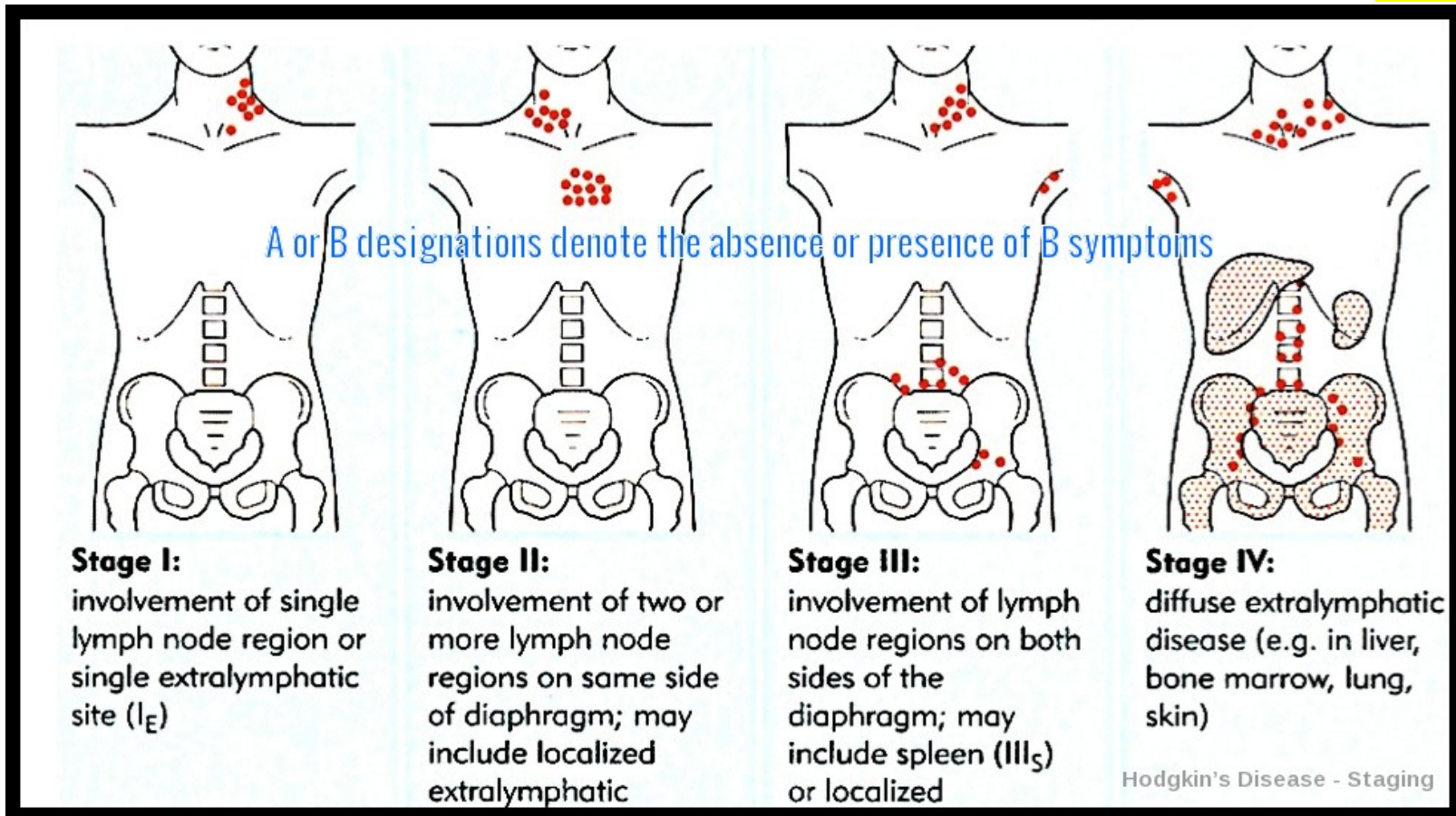
medicoapps.org/m-hodgkins-lymphoma/

Hodgkin's Lymphoma (HL)

- Physical Presentation
 - Fixed, firm, and painless enlarged lymph nodes
 - Neck, mediastinum, axillary, and groin region.
 - **B symptoms: Low-grade fever, weight loss, and night sweats**
 - Fatigue and pruritus
 - Involvement of mediastinum region
- Oral Presentation
 - Oral manifestations of Hodgkin's lymphoma are rare
 - **Rare**: Enlargement of the tonsillar tissue in Waldeyer's ring and paleness of the oral mucosa

Staging of Hodgkin's Lymphoma

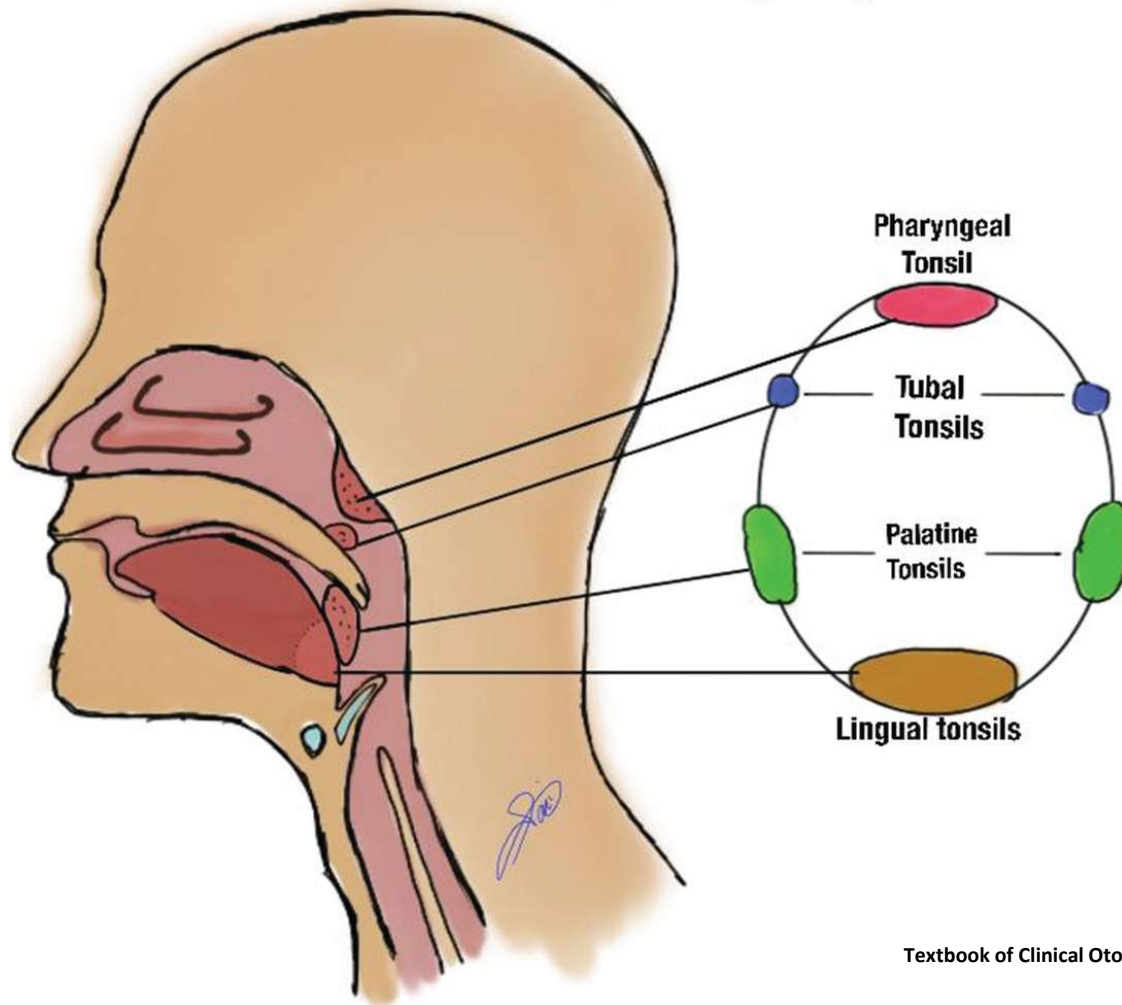
FYI



<https://askhematologist.com/hodgkins-disease/>

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Waldeyer ring of lymphoid tissues



Textbook of Clinical Otolaryngology

Table 23.7

WHO Classification of the Non-Hodgkin Lymphomas

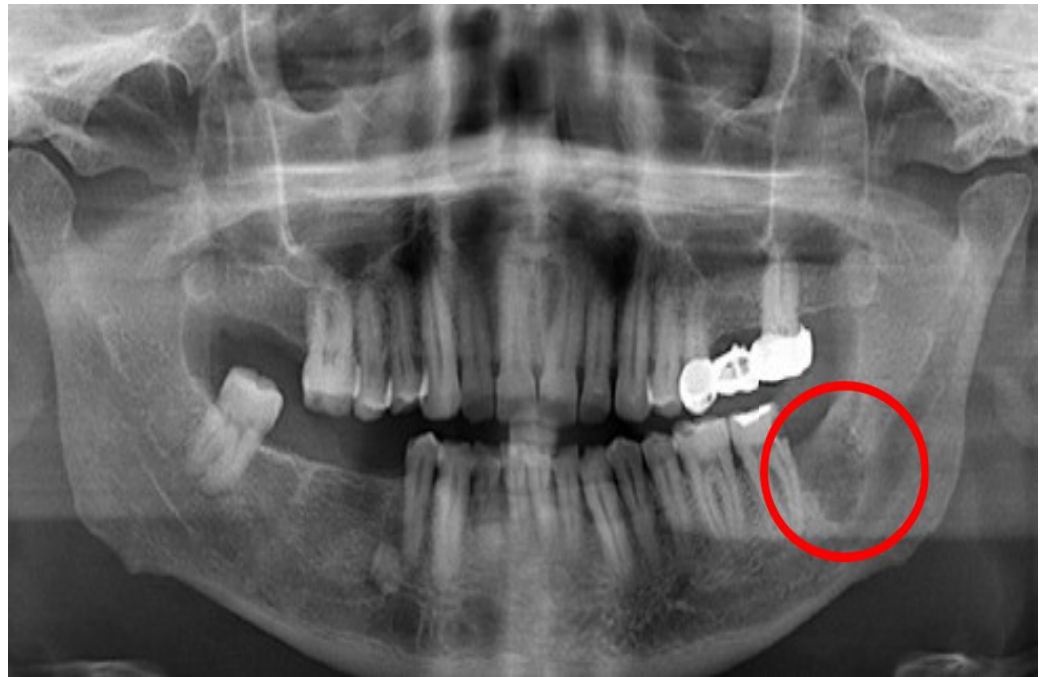
The Indolent Lymphomas
B-cell Neoplasms
Small lymphocytic lymphoma/B-cell chronic lymphocytic leukemia
Lymphoplasmacytic lymphoma ($\pm$ Waldenstrom's macroglobulinemia)
Plasma cell myeloma/plasmacytoma
Hairy cell leukemia
Follicular lymphoma (grade I and II)
Marginal zone B-cell lymphoma
Mantle cell lymphoma
T-cell Neoplasms
T-cell large granular lymphocyte leukemia
Mycosis fungoides
T-cell prolymphocytic leukemia
Natural Killer Cell Neoplasms
Natural killer cell large granular lymphocyte leukemia
The Aggressive Lymphomas
B-cell neoplasms
Follicular lymphoma (grade III)
Diffuse large B-cell lymphoma
Mantle cell lymphoma
T-cell Neoplasms
Peripheral T-cell lymphoma
Anaplastic large cell lymphoma, T/null cell
The Highly Aggressive Lymphomas
B-cell neoplasms
Burkitt's lymphoma
Precursor B lymphoblastic leukemia/lymphoma
T-cell neoplasms
Adult T-cell lymphoma/leukemia
Precursor T lymphoblastic leukemia/lymphoma

Source: Adapted from UpToDate, December 2021.

Non-Hodgkin's Lymphoma (NHL)

- 85% of the lymphomas diagnosed fall into this category
- Adults are mostly affected, but some cases of Non-Hodgkin's lymphoma can be seen in children.
- Physical Presentation
 - Enlarged lymph nodes and other lymphoid organs
 - B-symptoms
 - Extranodal presentation
- Oral Presentation
 - Swelling of the soft tissue of the oral mucosa
 - Enlargement of the minor or minor salivary glands
 - **Osteolytic (radiolucent)** lesion in the jaw bones

Non-Hodgkin's Lymphoma (NHL)



Iran J Med Sci. 2017 Jul; 42(4): 407–411.

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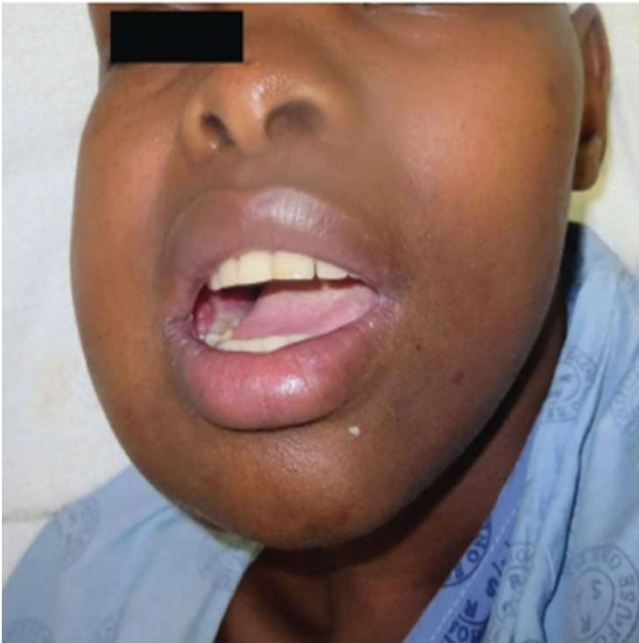
Burkitt's Lymphoma

- Aggressive type of Non-Hodgkin's, B-cell lymphoma
- Molecular abnormality: t(8;14)(q24;q32)
- Types
 - **Endemic**
 - Children of Central African descent
 - Sporadic
 - Older children and adults in Western countries
 - Immunodeficiency associated
 - Immunosuppressed individuals

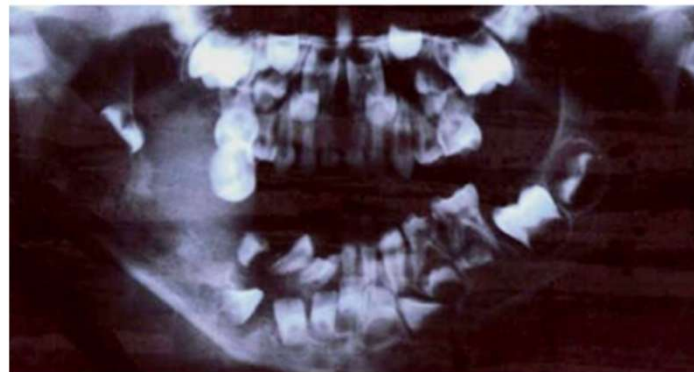
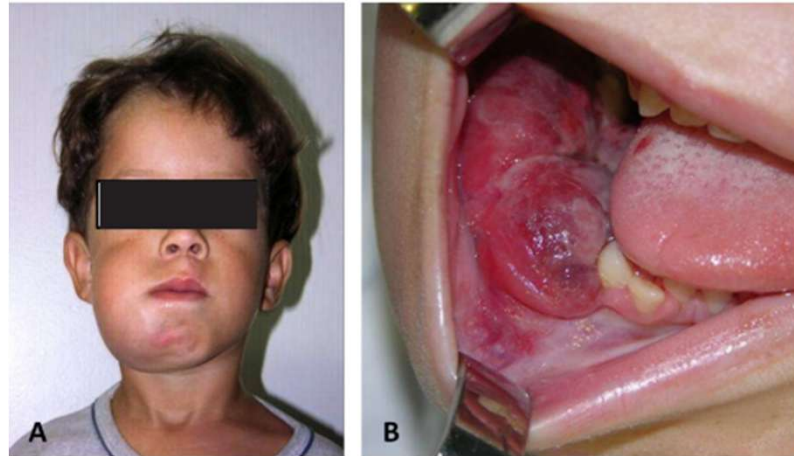
Burkitt's Lymphoma

- Endemic Type
 - Clinical Presentation
 - Rapidly growing tumor with a predilection for the abdominal organs
 - Abdominal tumors present as large masses accompanied by pain and vomiting
 - Oral Presentation
 - Large tumor of the maxilla or mandible exhibiting pain or paresthesia
 - Displacement of the teeth and can involve the oral soft tissues
 - Radiographs show poorly-defined osteolytic lesions

Burkitt's Lymphoma



S. Afr. dent. j. vol.70 n.4 Johannesburg May. 2015



Rev. odonto ciênc. 2010;25(4):417-421

Acute Lymphocytic Leukemia (ALL)

- More common in young children
- Physical Presentation
 - Enlargement of the lymphoid organs
 - Pallor, fever, fatigue, bone pain
 - Excessive bleeding and anemia
- Oral Presentation
 - Oral ulceration, infections, and paresthesia

Acute Myeloid Leukemia (AML)

- Form of leukemia seen in adult patients
- Proliferation of myeloblast
- Physical Presentation
 - Pallor, bone pain, and flu-like symptoms
 - Anemia, petechiae and bruising of the skin
 - Enlargement of the lymph nodes, tonsillar tissue and spleen
 - Poor wound healing and recurrent infections



www.everydayhealth.com/leukemia/rash-pictures-signs-symptoms/

Acute Myeloid Leukemia

- Oral Presentation
 - Submucosal hemorrhage, mucosal ulceration, severe periodontitis
 - **Gingival hyperplasia**
 - Oral microbial infections
 - Paresthesia



Santos, Paulo Sérgio da Silva et al. "Gingival leukemic infiltration as the first manifestation of acute myeloid leukemia." *Otolaryngology- Head and Neck Surgery* 143 (2010): 465 - 466.

Chronic Lymphocytic Leukemia (CLL)

- Tumor formed by partially mature clonal B lymphocytes
- Most common form of leukemia in adults
- Slow lymphocytic infiltrate of the bone marrow
- Physical Presentation
 - B-symptoms
 - **Lymphadenopathy**
 - Bleeding tendency, anemia, and slow wound healing
- Oral Presentation
 - Pallor throughout the oral cavity
 - Infection of the oral soft tissues



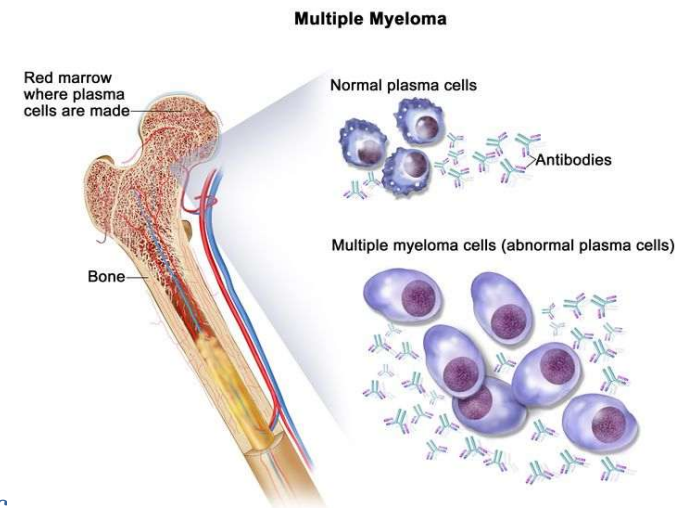
Haematology - Chronic leukaemia By Dr David Bareford and Dr Sebastian Francis on the 24 February 2011
(Photograph: SPL)

Chronic Myeloid Leukemia (CML)

- Neoplasm of partially mature myeloid cells
- Affects older adults
- **t(9;22) Philadelphia Chromosome [BCR-ABL]**
- Chronic, accelerated, and blast phases of disease
- Physical Presentation
 - B-symptoms, fatigue, excessive bleeding, and infection
 - Enlarged abdomen (splenomegaly)
 - Bleeding, anemia, and infection
- Oral Presentation
 - Enlargement of the lymphoid tissue, pallor, and infection

Multiple Myeloma

- Disease affecting the plasma cells
 - Clonal plasma cells
 - Abnormal cells secrete high amounts of protein products
 - **M-protein**
 - Bence Jones proteins
- Primarily seen in adults over the age of 40
- Plasmacytoma



Multiple Myeloma

- Physical Presentation
 - Hypercalcemia, renal failure, anemia, amyloidosis, and lytic bone lesions
 - Back pain, headaches, and fever
 - Increase risk of excessive bleeding and infection
- Oral Manifestations
 - Enlargement of the soft tissue
 - Lytic lesions of the jaws bone
 - Tooth loss

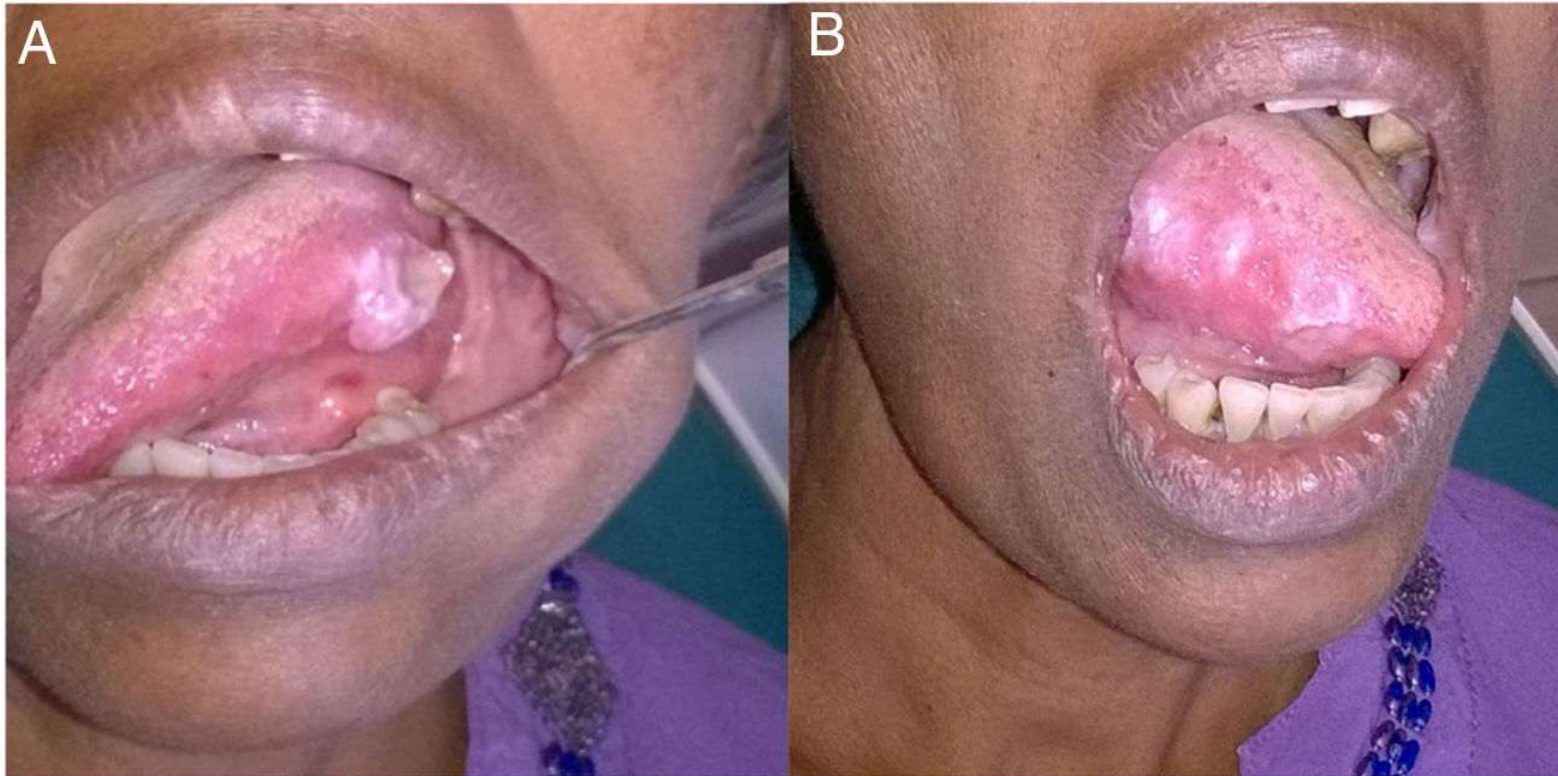
Amyloid Deposition with Multiple Myeloma



N Engl J Med 2011;365:71

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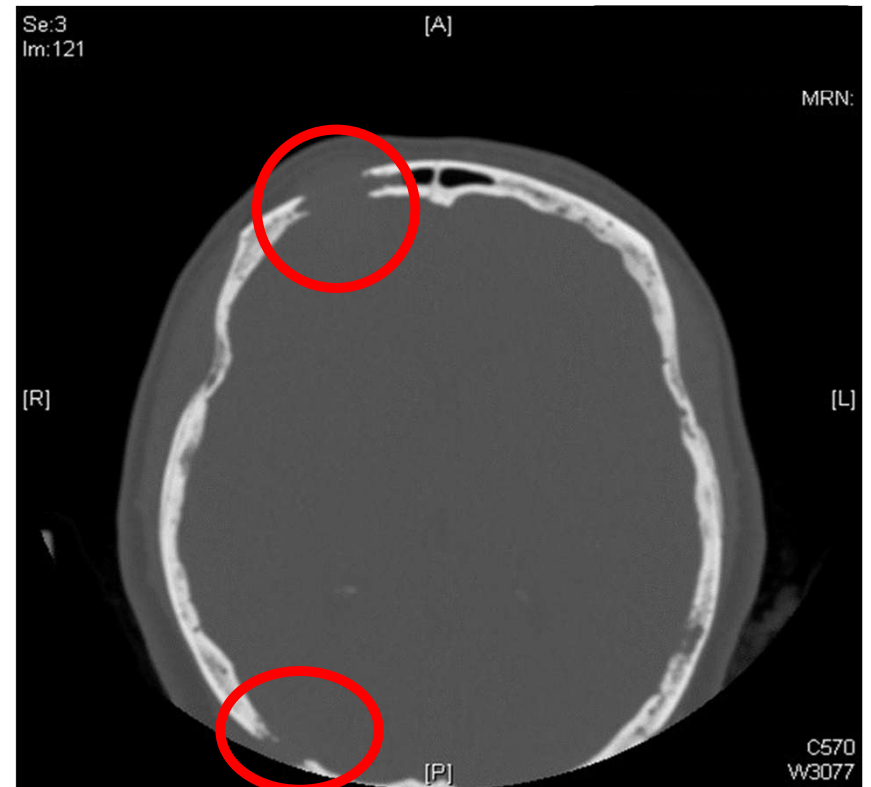
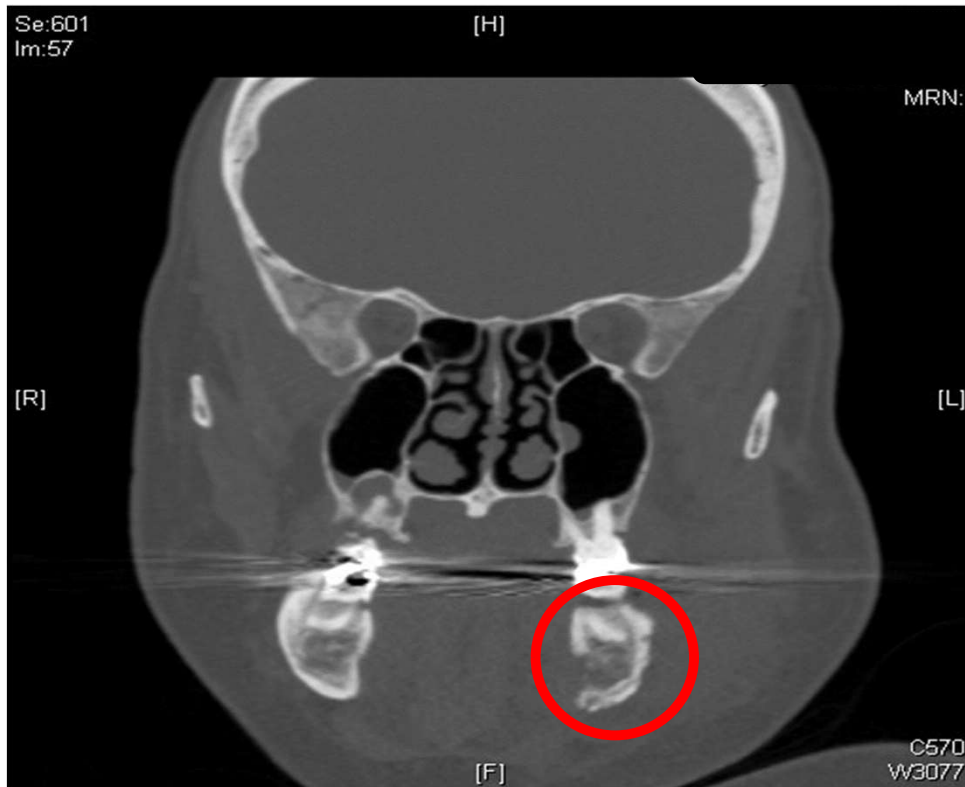
Oral Amyloid Deposition - Multiple Myeloma



Brazilian Journal of Otorhinolaryngology 22; 88(1): 146-149

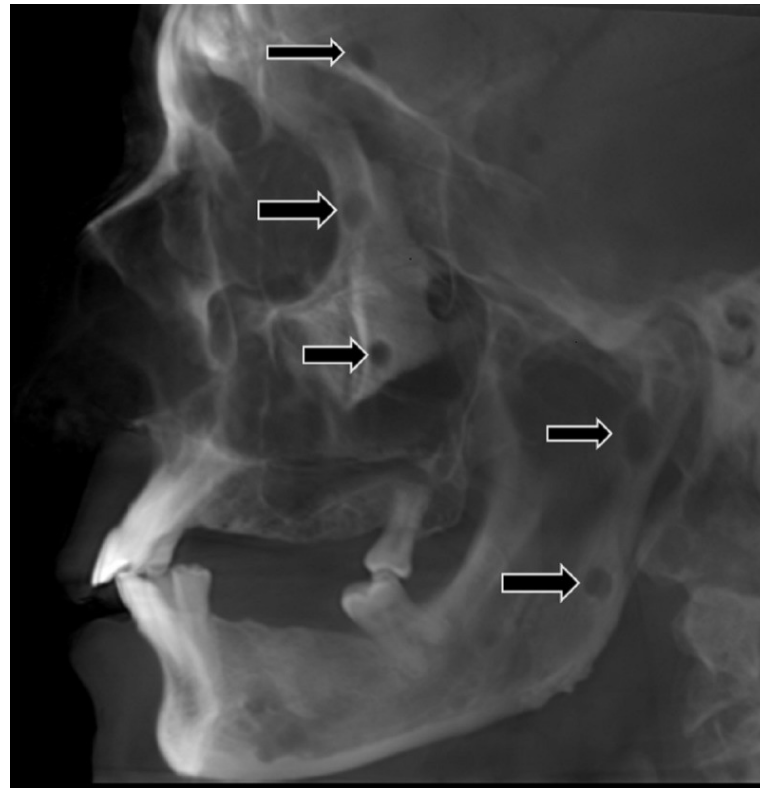
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Multiple Myeloma



Lytic bone lesions involving the mandible and skull “punched-out” radiolucencies
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Lateral Cephalogram – Multiple Myeloma



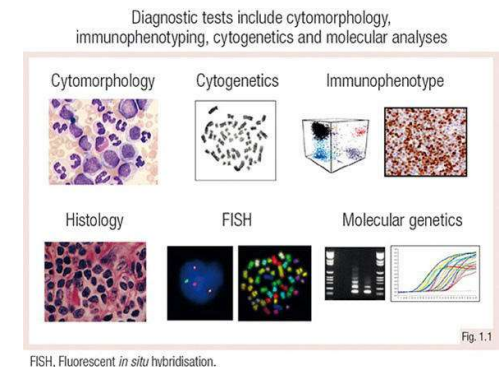
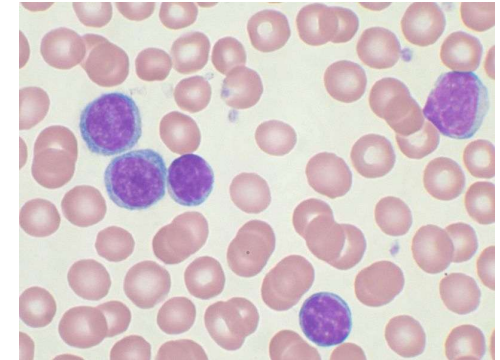
Cureus. 2018 Mar 3;10(3):e2265.

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Diagnosis and Treatment

- **Diagnostic modalities**

- Physical examination
- Medical history
- CBC with Differential
- Diagnostic imaging
- Blood and urine analysis
- Peripheral Blood smear
- Biopsy
- Immunophenotyping
- Cytogenetic testing



	Prevention of infection	Pain control	Maintain oral function	Managing oral complications of the underlying cancer or anti-cancer treatment	Quality of life
Before HSCT/CT	<u>Oral/Dental examination</u> - Refer for dental evaluation - Focus: Eliminate foci of infection, traumatic surfaces	<u>Prevention of local trauma</u> - Refer for dental evaluation - Eliminate causes for local trauma	<u>Chewing capability</u> - Ask patient about difficulty to chew - Refer to dentist to restore occlusion, if applicable	<u>Oral examination</u> Refer for an oral medicine specialist / dentist	<u>Education</u> Inform about future possible oral complications
During HSCT/CT	<u>Oral hygiene</u> - Ensure cleaning teeth and tongue - soft bristled toothbrush - flossing if capable w/o trauma <u>Decontamination</u> - Advise using bland rinses (e.g., saline), repeated rinses/day - Advise rinsing with chlorhexidine in alcohol free solution, x2/day - If unable to rinse, apply solution to sterile gauze or toothette <u>Prophylaxis</u> - Apply the institute protocol for prophylaxis for oral candidiasis & viral reactivation	<u>Pain assessment</u> - Score consistently (e.g., 0-10 scale/ WHO ladder) <u>Diagnosis</u> - Consult with an oral medicine / dentist for source of pain <u>Pain of dental origin</u> - Address using a non-surgical approach <u>Treatment</u> - Apply the institute protocol for pain control, topical & systemic agents	<u>Speaking: Oral moisture</u> - Utilize: - sugar free chewing gum/candy (non-acidic) - dentifrices for oral dryness - saliva substitute 1st line - frequent sips of water <u>Diet</u> - Promote non-cariogenic, low acid atraumatic diet	<u>Detection</u> - Detect possible signs & symptoms - Consult with oral medicine specialist / dentist treatment	<u>Taste change</u> - Encourage the patient to maintain oral intake <u>Dry Mouth</u> - Utilize: - sugar free chewing gum/candy (non-acidic) - dentifrices for oral dryness - saliva substitute - frequent sips of water <u>Awareness of future dental problems</u> - Educate patient regarding late effects of therapy
After HSCT/CT	<u>Viral infection</u> - Consider anti-viral prophylaxis <u>Candidal infection</u> - Consider anti-fungal prophylaxis: topical (sugar-free) or systemic <u>Dental infection</u> - Encourage the patient to have frequent dental recalls <u>Oral hygiene</u> - Educate & reinforce optimal oral hygiene and dental prevention	<u>Diagnosis</u> Consult oral medicine / dentist for source of pain	<u>Timing</u> Consult physician / oral medicine specialist / dentist for best timing for routine dental treatment & any prophylactic measures	<u>Dry Mouth</u> - Maintain good oral hygiene - use fluoride supplements - avoid sweets - visit the dentist regularly <u>Limited mouth opening</u> - Consider physiotherapy - Motivate routine dental care - Encourage oral hygiene <u>cGVHD (mucosa, salivary gland)</u> - Consult with an oral medicine specialist / dentist for current treatment recommendations - Motivate patient to keep routine surveillance (risk for SCC)	<u>Dry Mouth</u> - Motivate patients to: - keep oral hygiene - use fluoride supplements - avoid sweets - visit the dentist regularly - Utilize: - sialagogues - sugar free chewing gum/candy (non-acidic) - saliva substitute for palliation - frequent sips of water

Dental Considerations Prior to Cancer Therapy

- Pre-Treatment
 1. Risk of infection
 2. Exposure to infectious diseases
 3. Bleeding tendency/anemia
 4. Eradication of oral disease prior to cancer therapy
 5. Patient education

Dental Considerations During Cancer Therapy

- During Treatment
 1. Risk of infection
 2. Exposure to infectious diseases
 3. Bleeding tendency/anemia
 4. Eradication of oral disease prior to cancer therapy
 5. Patient education
 - 6. Side effects associated with cancer therapy**
 - 7. Coordinating dental treatment with cancer therapy**

Management for Hematologic Cancers

- Chemotherapy
- Radiation
- Stem cell transplantation
- Surgery
- Prednisone – often used in combination with other type of therapy
- Biologic therapy
 - Immunotherapy
 - Targeted therapy
- Bisphosphonates*

Side effects of Cancer Therapy



Radiation-Induced Caries



Oral Mucositis



Fungal Infection/Candidiasis/ Oral Thrush

Dental Considerations During Remission

- Remission
 1. Permanent changes to the oral cavity and overall health after cancer treatment.
 1. Trismus
 2. Growth and developmental changes in children
 2. Side effects of post cancer treatment therapy
 1. Graft vs Host Disease
 2. Radiation-Induced Osteonecrosis of the Jaw (**ORN**)
 3. Medication-Related osteonecrosis of the Jaw (**MRONJ**)

Trismus



MRONJ



Oral and Maxillofacial Pathology, 4ed.

Resources

